Protein Foundation Models for Predicting Interaction Signals in Biomolecular Condensate Assemblies

Xingyuan Zhang (senecio@stanford.edu)
Department of Chemistry, Stanford University

Pengwei Sun (pengwei@stanford.edu)
Departments of Radiology and Radiation Oncology, Stanford University

Abstract

We study whether sequence foundation models can provide useful interaction signals for retrieving components of a biomolecular condensate assembly. Starting from a human STRING experimental interaction graph (19.7k nodes), we build FM-only weighted graphs where edges encode interaction propensity proxies derived from masked-language-model behavior (pseudo-log-likelihood difference, perturbation coupling, and cross-block attention). We then fuse FM signals with STRING experimental edges via reweighting and gated mixing, and rank candidate proteins from four experimentally motivated seed proteins using personalized PageRank. Validation uses a lab-generated condensate proteomics-MS dataset: high-confidence proteins from a volcano-plot threshold define positives. Across graphs, STRING experimental is a strong baseline; FM-only propensity graphs recover a substantial fraction of positives; and FM-guided reweighting of STRING yields the best precision and enrichment in our benchmark.

1 Problem Setting and Benchmark

Biomolecular condensates form dynamic assemblies whose component proteins are often identified via proteomics enrichment. Given a small set of known "core" proteins, the goal is to retrieve additional assembly components by diffusing information over a protein interaction network.

We benchmark on condensation-induced Src kinase activation, using four seeds (SRC1, FNBP1, FNBP1L, TRIP10) and validating against a lab-built condensate MS interactome.

2 Related Work and Baselines

Protein foundation models. Transformer-based protein language models (e.g., ESM-2, ProtT5) trained on large protein corpora have become general-purpose sequence representation models and have demonstrated utility on diverse proxy tasks such as remote homology, contact prediction, and function-related inference.

PPI databases and experimental baselines. STRING aggregates interaction evidence from multiple sources. In this project, we use the STRING experimental subnetwork as an external experimental baseline graph.

Graph diffusion for biological retrieval. Personalized PageRank / random-walk-with-restart is widely used to propagate seed information in biological networks to rank candidate genes or proteins.

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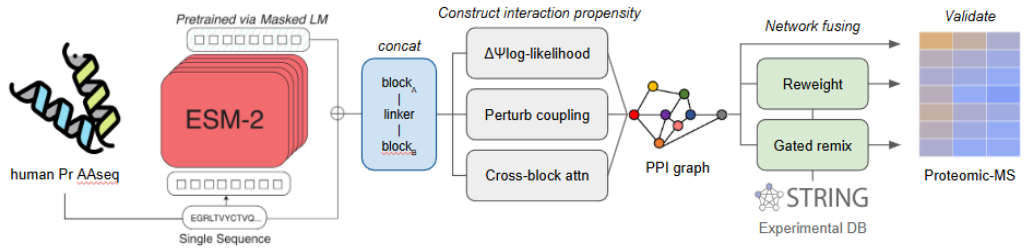

Figure 1: Project pipeline: protein FM embeddings → FM-derived interaction propensity proxies → network fusion with STRING experimental edges → validation with condensate proteomic-MS.

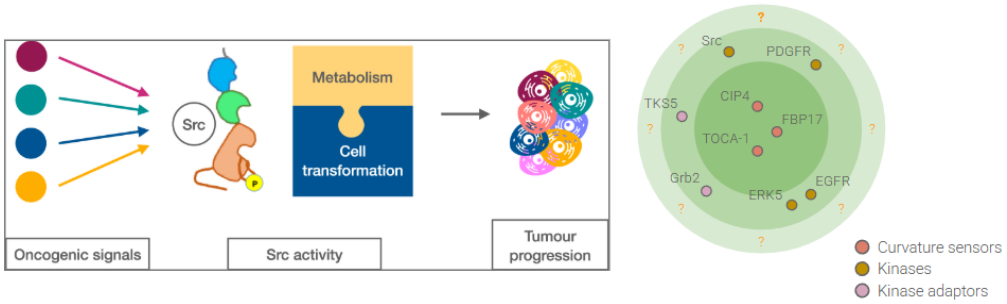

Figure 2: Benchmark task: condensation-induced Src kinase activation. Seeds diffuse on each graph to retrieve assembly components; validated with lab-built condensate MS interactome.

We compare diffusion on

- STRING experimental
- FM similarity kNN graphs
- FM propensity-proxy graphs
- fused graphs produced by reweighting and gated mixing

3 Methods

3.1 FM embeddings

We embed the human protein universe (19k nodes) using ESM2 and/or ProtT5 to obtain a vector representation for each protein sequence.

3.2 FM-only interaction propensity graphs

We construct FM-derived graphs where weighted edges represent interaction propensity proxies computed without external interaction labels.

We evaluate three proxies:

- ΔPLL (pseudo-log-likelihood difference)
- Perturb coupling
- Cross-block attention

Candidate pairs are generated via a blocking step (top-M neighbors in embedding space) to reduce compute; final edge weights come from proxy scores rather than embedding similarity.

3.3 Fusion with STRING experimental

We fuse FM signals with STRING experimental edges using two operators.

Reweight:

Scale each STRING edge weight by a monotone function of the FM proxy on that edge.

Gated mix:

Gate STRING edges using FM proxy and mix gated STRING edges with FM-only edges through a convex combination.

These operators treat FM proxies as a calibration signal rather than replacing experimental evidence.

3.4 Diffusion

For each graph, we run personalized PageRank with a personalization vector concentrated on the four seeds, and rank all other nodes by their stationary probability.

4 Experimental Setup

4.1 Proteomics-MS positives from volcano plot

We define positives using our condensate proteomics-MS assay.

A volcano plot threshold using enrichment effect size and statistical significance (e.g., q/FDR) yields a high-confidence set of proteins enriched in the condensate.

After excluding the four seeds, we obtain 543 positives used for evaluation.

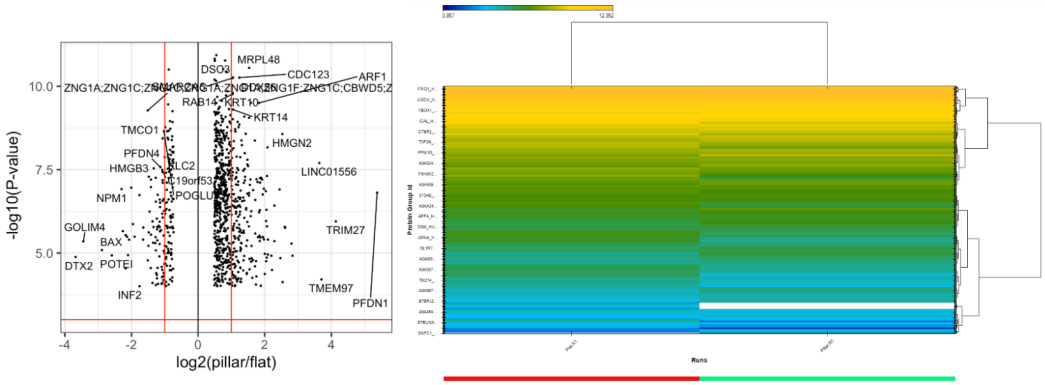

Figure 3: Volcano plot and enrichment benchmark of proteomic MS.

4.2 Metrics

We report

- AUPRC
- Precision@k and Recall@k (k = 200)
- Enrichment Factor@k

All graphs are evaluated on the same node universe and the diffusion parameter α is fixed across comparisons.

5 Results and Analysis

5.1 Quantitative results

Key trends observed:

- STRING experimental is a strong baseline. Diffusion on STRING yields high early retrieval of MS positives.
- FM-only propensity graphs are informative. PLL-based proxies substantially outperform FM similarity kNN baselines.
- FM-guided reweighting improves retrieval. Reweighting STRING edges with FM PLL yields the best overall performance.

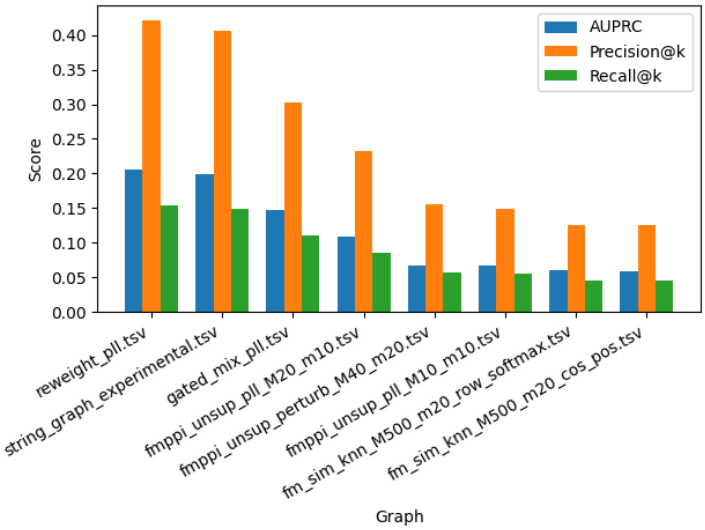

Figure 4: Retrieval performance across graphs (AUPRC / Precision@k / Recall@k).

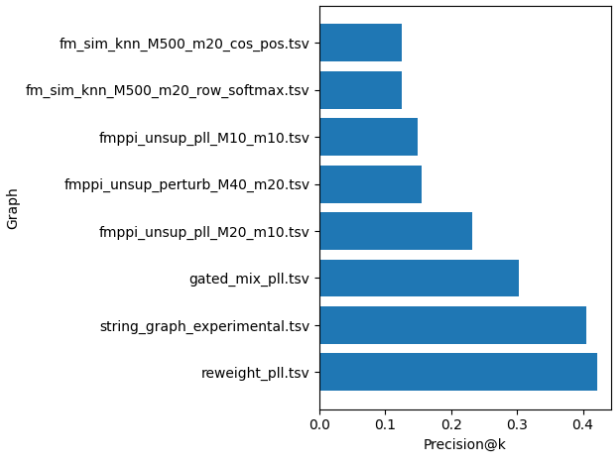

Figure 5: Precision@k comparison (k fixed; higher is better).

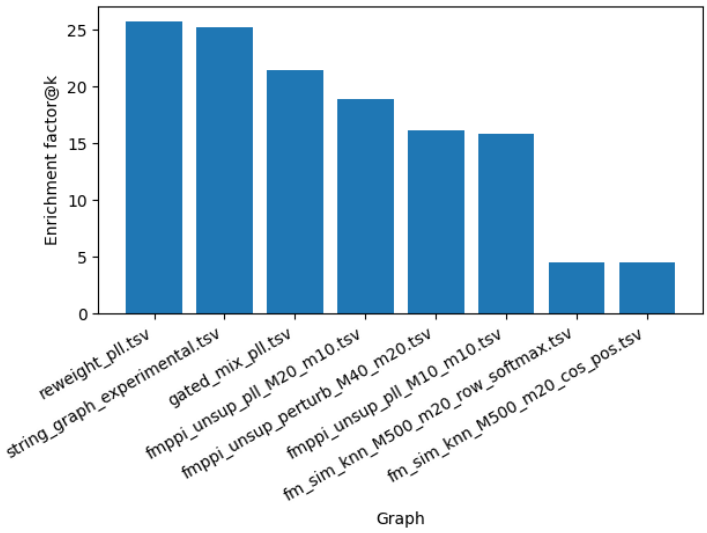

Figure 6: Enrichment Factor@k across graphs.

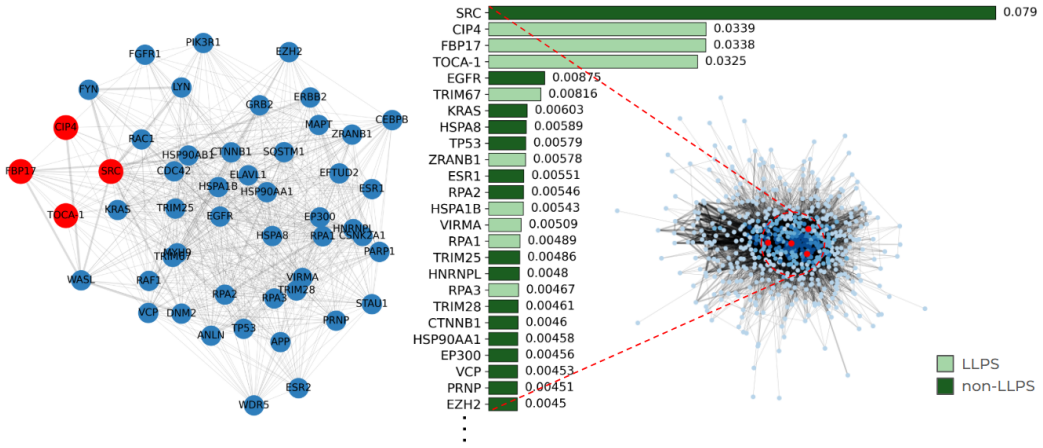

Figure 7: Graph diffusion results.

5.2 Qualitative interpretation

FM similarity kNN graphs tend to retrieve proteins that are sequence-similar to seeds but are not necessarily relevant to the condensate assembly.

In contrast, masked-LM-derived propensity signals appear to capture non-trivial compatibility information relevant to assembly membership.

Fusion through reweighting preserves the experimental network structure while amplifying edges supported by FM signals.

6 Novelty, Rigor, and Limitations

Novelty

We propose an unsupervised construction of FM-derived interaction propensity graphs (PLL and perturb coupling) and use them as calibration signals for modulating experimental interaction networks.

Rigor

We compare multiple graph families under a fixed diffusion procedure and validate results using experimentally derived MS positives.

Limitations

- FM proxies do not necessarily represent direct physical binding.
- Candidate-pair blocking introduces sampling bias.
- Evaluation is performed on a single benchmark system.
- Hyperparameter sweeps were limited.

7 Future Work

Future directions include

- bootstrap stability analyses
- extension to multiple condensate systems
- improved FM-only graph construction
- learned fusion using independent PPI labels

8 Contributions

- X.Z. proposed the project overview, conducted the construction of pipelines and model training, wrote the descriptions and results. P.S. conducted the construction of pipelines and model training, wrote the descriptions and results.

9 Code Availability

- All source code and pipelines could be found in this GitHub repository: https://github.com/xing-yuan-zhang/plm-geometry

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